Model Selection: MAP, MLE

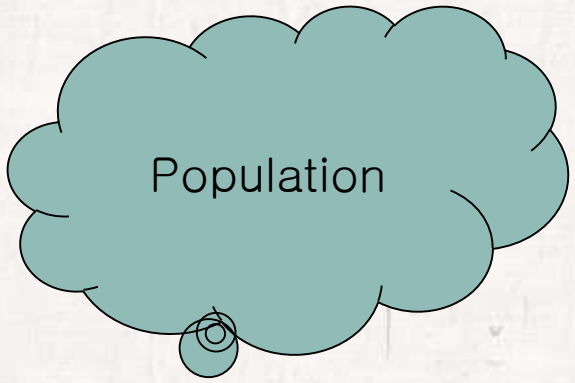
Introduction

- How to find a model for given data
  - Based on Error Function
    - Define an error function
    - Find a model which minimizes the error function
      - Solving linear equation
      - Gradient descent method
  - Based on Probability
    - Evaluate the probability of given data generated
    - Find a model which maximize the probability
      - Solving linear equation
      - Gradient descent method
      - Estimation-Maximization
      - MCMC (Makov Chain Monte Carlo)

Model Selection

- You have sample data $X = (\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n)$
  - do not know about population.
  - want to estimate some hidden parameters of population

$X = (\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n)$ ⟵



Estimate Parameters, Assumption,
Models, or Any Information of Population
based on the given samples

HOW?

Model Selection

- Let's assume that we have
  - a set of sample data
$$D = \{d_1, d_2, \cdots, d_n\}$$
  - a set of candidate models, parameters or assumptions
$$M = \{m_1, m_2, \cdots, m_k\}$$

- Usually
  - $P(D \mid m_i)$ is easy to evaluate, but $P(m_i \mid D)$ is hard
  - Why?

- Questions
  - Choose the most probable model given the data

Model Selection

- Choose a model by Maximum A Posteriori (MAP)
  - Evaluate the conditional probability of each *m* given *D*

$$P(m_1 \mid D), P(m_2 \mid D), \cdots, P(m_k \mid D)$$

  - Choose the model which has the maximum probability
  - In Short

$$\text{model} = \arg\max_{m \in M} P(m \mid D)$$

Model Selection

- Choose a model by Maximum A Posteriori (MAP)

$$\text{model} = \arg\max_{m \in M} P(m \mid D)$$

$$= \arg\max_{m \in M} \frac{P(D \mid m)P(m)}{P(D)}$$

$$= \arg\max \left\{ \frac{P(D \mid m_1)P(m_1)}{P(D)}, \cdots, \frac{P(D \mid m_k)P(m_k)}{P(D)} \right\}$$

$$= \arg\max_{m \in M} P(D \mid m)P(m)$$

→ Usually, it is hard to obtain

→ Usually, it is easy to obtain

- Also, remember

$$\arg\max_{m \in M} P(m \mid D) = \arg\max_{m \in M} \log P(m \mid D)$$

Model Selection

- Maximum Likelihood Estimator (MLE)

$$\text{model} = \arg\max_{m \in M} P(D \mid m) P(m)$$

Assume that *P(m)* is the same for all *m*

$$= \arg\max_{m \in M} P(D \mid m)$$

Probability of *D*, or Likelihood of *m*

  - Choose a model which maximizes the likelihood

$$\text{model} = \arg\max_{m \in M} P(D \mid m)$$

Model Selection

- Maximum Likelihood Estimator (MLE)
  - Evaluate the conditional probability of *D* given each of *m*

$$P(D \mid m_1), P(D \mid m_2), \cdots, P(D \mid m_k)$$

  - Choose the model which has the maximum probability
  - In Short

$$\text{model} = \arg\max_{m \in M} P(D \mid m)$$

Example-Coin Tossing

- Example: Coin Tossing
  - We have five coins in a pockets, and choose one randomly and toss it
    - One of c_1: two heads
    - Three of c_2: one head and one tail
    - One of c_3: two tails
  - Now, we see a head. Which coin do you chose?
    - What is the Sample? $D = \{h\}$
    - What are the candidates of what to estimate? $M = \{c_1, c_2, c_3\}$
  - MAP estimator of coin

$$\begin{aligned} \text{model} &= \arg\max_{c \in \{c_1, c_2, c_3\}} P(c \mid D) \\ &= \arg\max_{c \in \{c_1, c_2, c_3\}} P(c \mid h) \\ &= \arg\max_{c \in \{c_1, c_2, c_3\}} P(h \mid c) P(c) \end{aligned}$$

Example-Coin Tossing

- Example: Coin Tossing

$$\text{model} = \arg\max_{c \in \{c_1, c_2, c_3\}} P(c \mid h) P(c)$$

  - We know that

$$P(c_1) = 0.2, P(c_2) = 0.6, P(c_3) = 0.2$$

$$P(h \mid c_1) = 1.0, P(h \mid c_2) = 0.5, P(h \mid c_3) = 0.0,$$

  - Thus,

$$\left.\begin{aligned} P(h \mid c_1)P(c_1) &= 0.2 \\ P(h \mid c_2)P(c_2) &= 0.3 \\ P(h \mid c_3)P(c_3) &= 0.0 \end{aligned}\right\} \xrightarrow{\max} P(h \mid c_2)P(c_2) = 0.3$$

MAP estimator is c_3

Linear Regression

- We have *n* sample data

$$D = \{D_1, D_2, \cdots, D_n\} \text{ where } D_i = (\mathbf{x}_i, y_i)$$

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \cdots, x_{id}) \text{ is an input in } d \text{ dimensional space}$$

$$y_i \text{ is the output for } \mathbf{x}_i$$

- We want to find a line *f(x)* which best fits *D*
  - What does "best fit" means?

$$\text{Find } f(\mathbf{x}) \text{ which minimizes error, } \sum_{i=1}^{n} (f(\mathbf{x}_i) - y_i)^2$$

Linear Regression

- What does "best fit" means?
  - New Interpretation

Find $f(\mathbf{x})$ which maximizes the probabilit y to observe D

Find $f(\mathbf{x})$ which maximizes $p(D \mid f(\mathbf{x}))$

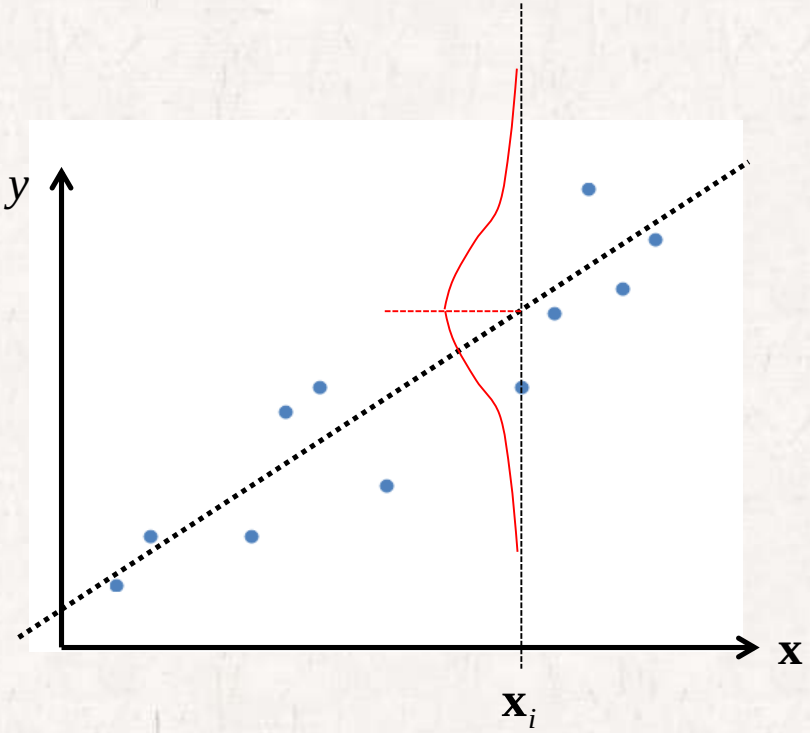
  - Why not?

Find $f(\mathbf{x})$ which has the maximum probabilit y given D

Find $f(\mathbf{x})$ which maximizes $p(f(\mathbf{x}) \mid D)$

MLE of Linear Regression

- Assumption for a probabilistic model
  - The data is generated from a linear model f(x)
  - However, a random noise, ε, from $N(0,\sigma^2)$ is added to data

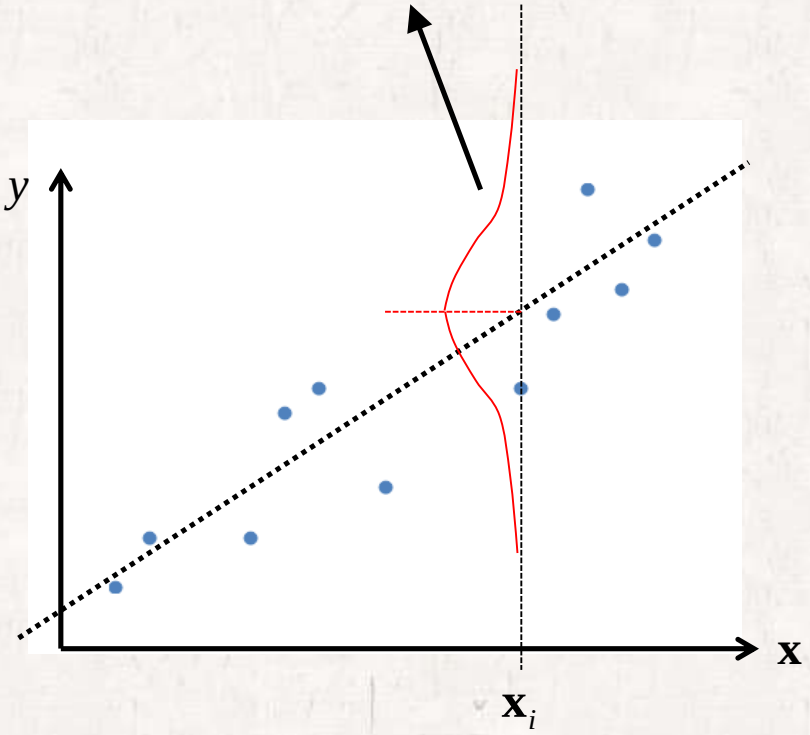


MLE of Linear Regression

- Probabilit y that y_i is observed when $\mathbf{x}_i$, $f(\mathbf{x})$ and σ are given

$$P(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2} \cdot (y_i - f(\mathbf{x}_i))^2 \right)$$

  - where $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$



MLE of Linear Regression

- You have data

$$\mathbf{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \cdots, (\mathbf{x}_n, y_n)\}$$

- Probabilit y of $\mathbf{D}$ when $f(\mathbf{x})$ and σ are given

$$P(\mathbf{D} \mid \mathbf{w}, \sigma) = \prod_{i=1}^{n} P(y_i, \mathbf{x}_i \mid \mathbf{w}, \sigma)$$

Likelihood of $\mathbf{w}$

  - $\mathbf{x}_i$ and $\mathbf{x}_j$ are independen t
  - $\mathbf{x}_i$ is independen t to $\mathbf{w}, \sigma$

MLE of Linear Regression

- Probabilit y of **D** when $f(\mathbf{x})$ and σ are given

$$P(\mathbf{D} \mid \mathbf{w}, \sigma) = \prod_{i=1}^{n} P(y_i, \mathbf{x}_i \mid \mathbf{w}, \sigma)$$

Assume that it is a constant

$$= \prod_{i=1}^{n} P(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma) P(\mathbf{x}_i \mid \mathbf{w}, \sigma) = \prod_{i=1}^{n} P(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma) P(\mathbf{x}_i)$$

$$\mathbf{w} = \arg\max_{\mathbf{w}} P(\mathbf{D} \mid \mathbf{w}, \sigma) = \arg\max_{\mathbf{w}} \log P(\mathbf{D} \mid \mathbf{w}, \sigma)$$

$$= \arg\max_{\mathbf{w}} \log \prod_{i=1}^{n} P(y_i, \mathbf{x}_i \mid \mathbf{w}, \sigma)$$

$$= \arg\max_{\mathbf{w}} \log \prod_{i=1}^{n} P(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma) p(\mathbf{x}_i \mid \mathbf{w}, \sigma)$$

$$= \arg\max_{\mathbf{w}} \log \prod_{i=1}^{n} P(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma)$$

MLE of Linear Regression

$$
\begin{aligned}
\mathbf{w} &= \arg\max_{\mathbf{w}} P(\mathbf{D} \mid \mathbf{w}, \sigma) \\
&= \arg\max_{\mathbf{w}} \log \prod_{i=1}^{n} P(y_i \mid \mathbf{x}_i, \mathbf{w}, \sigma) \\
&= \arg\max_{\mathbf{w}} \log \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right) \\
&= \arg\max_{\mathbf{w}} \sum_{i=1}^{n} \log\left(\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right)\right) \\
&= \arg\max_{\mathbf{w}} \sum_{i=1}^{n} \left(\log\frac{1}{\sqrt{2\pi}\sigma} + \log\left(\exp\left(-\frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right)\right)\right) \\
&= \arg\max_{\mathbf{w}} \sum_{i=1}^{n} \left(\log\frac{1}{\sqrt{2\pi}\sigma} - \frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right) \\
&= \arg\max_{\mathbf{w}} \left(\sum_{i=1}^{n} \log\frac{1}{\sqrt{2\pi}\sigma} - \sum_{i=1}^{n} \frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right) \\
&= \arg\max_{\mathbf{w}} \left(-\sum_{i=1}^{n} \frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right) \\
&= \arg\min_{\mathbf{w}} \left(\sum_{i=1}^{n} \frac{(y_i - f(\mathbf{x}_i))^2}{2\sigma^2}\right) \\
&= \arg\min_{\mathbf{w}} \left(\sum_{i=1}^{n} (y_i - f(\mathbf{x}_i))^2\right)
\end{aligned}
$$

Find $\mathbf{w}$ which minimizes $\sum_{i=1}^{n} (y_i - f(\mathbf{x}_i))^2$

Equivalent to minimizing squared error functions

MLE solution and SLE solution for Linear Regression are the same

Summary

- You may choose a model which
  - minimizes the error
  - maximizes the probability